

Seyyed Arad Hosseini Moghaddam

COMPREHENSIVE AI / DATA SCIENCE KNOWLEDGE CV

spikick.cod@gmail.com | Portfolio | LinkedIn | GitHub | Hugging Face | Iran

Purpose: Source-of-truth profile document for the portfolio RAG chatbot. It is intentionally more detailed than a recruiter résumé and preserves verified technical facts, project metrics, tools, public credentials, and explicit scope boundaries.

IDENTITY AND PROFESSIONAL PROFILE

- Name: Seyyed Arad Hosseini Moghaddam.
- Primary professional identity: Data Scientist, AI Engineer, and Machine Learning Engineer.
- Location listed publicly on LinkedIn: Iran.
- Work context publicly listed on LinkedIn: Personal Projects.
- Strongest specialization stated on the portfolio: computer vision, especially medical imaging.
- Additional focus areas: deep learning, NLP, LLMs, RAG, agentic AI, model fine-tuning, applied analytics, and MLOps.
- Works across the ML lifecycle: data preparation, experimentation, model design, training, evaluation, deployment, observability, and iteration.
- Publishes source code on GitHub and models, adapters, and datasets on Hugging Face.

TECHNICAL CAPABILITIES AND SKILLS

Data Science & ML

Python · SQL · pandas · NumPy · scikit-learn · Statistics · Data Analytics · Feature Engineering · Data Visualization

Deep Learning & Medical AI

PyTorch · TensorFlow · Computer Vision · Medical Imaging · CNNs · 3D Segmentation · NLP · Model Evaluation · MONAI · 3D U-Net · NIfTI · Mixed Precision (AMP) · Sliding-window inference

Generative AI & Fine-tuning

LLMs · Transformers · Hugging Face · Fine-tuning · LoRA · QLoRA · PEFT · Prompt Engineering · Embeddings · RAG · Vector Databases · Semantic Search · LangChain · BPE tokenization · Causal Language Modeling · 4-bit NF4 quantization

Agentic AI & Orchestration

Agentic AI · AI Agents · LangGraph · LangSmith · Deep Agents · ReAct Agents · Tool Calling · Multi-Agent Systems · Subgraphs · Stateful Agents · Conversational Memory · Human-in-the-Loop · LLM Observability · LLM Cost Optimization · MCP · ChromaDB

MLOps & Engineering

Git · FastAPI · MLflow · Docker · GitHub Actions · CI/CD · REST APIs · SSE · SQLite · Testing

Software & Web

HTML · CSS · JavaScript · Tailwind CSS · Django · PyQt6 · Gradio

DETAILED PROJECT RECORD

1. Persian GPT - 110.4M Transformer Trained From Scratch

NLP · Transformers · PyTorch · BPE · Causal LM · Persian

Built a decoder-only GPT-style Persian language model end to end with no pretrained weights, pretrained tokenizer, or distillation. Created the cleaned corpus, tokenizer, model, training loop, baselines, evaluation, and public Hugging Face release.

- Corpus: 855,724 cleaned Persian Wikipedia documents; 244M training tokens.
- Data preparation included Unicode normalization, Arabic-to-Persian character mapping, ZWNJ preservation, filtering, deduplication, and train/validation/test separation.
- Tokenizer: 32,000-token BPE trained from scratch; ZWNJ preserved.
- Architecture: 12-layer decoder-only transformer; hidden size 768; 12 attention heads; FFN 3072; context length 1024; learned absolute positional embeddings; Pre-LayerNorm; GELU; tied input/output embeddings.
- Training: AdamW, linear warmup (500 steps), cosine decay, peak LR 3e-4, effective batch 64, 2 epochs / 7,450 optimizer steps, mixed precision on a single NVIDIA A100.
- Evaluation: test perplexity 14.52; best validation loss 2.6391; test loss 2.6756.

- Baseline comparison: unigram perplexity 1590.26 and interpolated 5-gram perplexity 171.60; model improved 11.8x over the 5-gram baseline.
- Published the model and cleaned dataset on Hugging Face with reproducible loading and a public demo/workflow.
- Known limitation documented by the author: Wikipedia-only training produces encyclopedic rather than conversational Persian.

[Project / model source](#)

2. 3D Cardiac MRI Segmentation

[Medical AI](#) · [Computer Vision](#) · [PyTorch](#) · [MONAI](#) · [3D U-Net](#)

Developed an end-to-end voxel-level cardiac MRI segmentation pipeline using PyTorch, MONAI, and a 3D U-Net, including volumetric preprocessing, training, evaluation, sliding-window inference, and visualization.

- Dataset: Medical Segmentation Decathlon - Task02 Heart; NIFTI (.nii.gz) volumes and segmentation labels.
- Architecture: 3D U-Net with encoder-decoder structure, skip connections, 3D convolutions, and multi-scale feature extraction.
- Training: AdamW optimizer, DiceCELoss, 100 epochs, AMP/FP16 mixed precision.
- Inference: sliding-window inference for large 3D volumes.
- Evaluation: best validation Dice score 0.8821.
- Built preprocessing and visualization workflows around MONAI and volumetric medical data.
- Used metric analysis to identify why high raw accuracy can be misleading under segmentation class imbalance.

[Project / model source](#)

3. Messy2JSON - Qwen2.5-0.5B Structured Extraction

[LLM Fine-tuning](#) · [QLoRA](#) · [PEFT](#) · [4-bit NF4](#) · [Structured Output](#)

Fine-tuned Qwen2.5-0.5B-Instruct with QLoRA to convert messy English text into JSON, targeting lightweight local structured-information extraction.

- Base model: Qwen/Qwen2.5-0.5B-Instruct.
- Method: QLoRA with 4-bit NF4 quantization; LoRA rank 16, alpha 32, dropout 0.05.
- Trainable parameters: ~4M of ~498M (~0.85%).
- Dataset: standardlogic/data-extraction-sft-100k; 39k JSON-filtered samples, with 9,000 training and 1,000 validation samples used.
- Training: 2 epochs, learning rate 2e-4, max sequence length 1024; ~40 minutes on a T4 GPU.
- Results: final validation loss 0.0045; mean token accuracy 99.82%.
- Documented limitations include schema variability, occasional nesting quirks, English-only scope, and no guaranteed schema enforcement.

[Project / model source](#)

4. AI Research Assistant Platform

[Agentic AI](#) · [LangGraph](#) · [RAG](#) · [LangSmith](#) · [ChromaDB](#) · [FastAPI](#) · [Docker](#)

Built a production-style research assistant around a LangGraph multi-agent workflow, combining planning, retrieval/research, tool execution, synthesis, reflection, persistent state, observability, APIs, testing, and containerized deployment.

- Planner routes work across retrieval, research, and tool-execution branches; compatible branches can run in parallel.
- Five agent roles with synthesis and bounded reflection/refinement behavior.
- RAG layer uses ChromaDB; state persistence uses SQLite.
- LangSmith tracing/evaluation for observability and inspection.
- FastAPI exposes REST and server-sent events (SSE) interfaces.
- Docker-ready deployment and 103 offline tests documented on the portfolio.
- Emphasis on citation-grounded answers and controlled refinement rather than unbounded agent loops.

[Project / model source](#)

5. English-to-Persian Psychology Translation Model

[NLP](#) · [Translation](#) · [NLLB-200](#) · [Fine-tuning](#) · [English/Persian](#)

Fine-tuned facebook/nllb-200-distilled-600M for English-to-Persian translation focused on psychology, counseling, mental-health education, self-help, and related terminology.

- Domain focus: psychology articles, academic/educational mental-health material, counseling and therapy-related text, and self-help/personal-development content.
- Designed to improve handling of specialized terminology compared with general-purpose translation.
- Positioned as translation assistance rather than a substitute for professional translators or mental-health professionals.

PUBLIC MODELS, DATASETS, AND REPRODUCIBILITY

- Persian GPT model: aradmanamnaoon/persian-lm-from-scratch on Hugging Face.
- Persian training dataset: aradmanamnaoon/persian-wikipedia-cleaned-for-persian-NLP on Hugging Face.
- Messy2JSON merged model: aradmanamnaoon/messy2json-qwen-0.5b.
- Messy2JSON LoRA adapter is also published separately on Hugging Face.
- Medical-image segmentation and AI Research Assistant source code are published on GitHub.
- Public work emphasizes reproducible training details, explicit limitations, evaluation metrics, and inspectable source code.

CERTIFICATIONS

- AI Capstone Project with Deep Learning — Coursera, issued 2026.
- Machine Learning with Python (V2) — Coursera, issued 2026.
- Python Project for Data Science — Coursera, issued 2026.

EDUCATION

- Vocational High School — listed on LinkedIn for 1403-1406 Solar Hijri (approximately 2024-2027/28 Gregorian).
- No university degree is publicly listed in the sources used to compile this document.

LANGUAGES

- Persian: Native or bilingual proficiency.
- English: Professional working proficiency.

CURRENT AREAS OF EXPLORATION

- Deep learning for vision, language, and multimodal tasks.
- Medical AI and medical-image analysis.
- Applied analytics: converting messy data into insights, models, and measurable decisions.
- Generative AI and modern generative architectures.
- LLM fine-tuning with LoRA, QLoRA, PEFT, and domain-specific datasets.
- RAG systems with embeddings, semantic search, and vector databases.
- Agentic AI engineering with stateful tool-using workflows, LangGraph, LangSmith, MCP, and human-in-the-loop patterns.

VERIFIED QUESTION-ANSWER FACTS FOR RAG GROUNDING

Q: What kind of AI systems has Arad built?

A: Multiple agentic AI and RAG systems, tool-using workflows, retrieval pipelines, vector search, stateful interactions, LLM orchestration, deep-learning systems, and traditional machine-learning workflows.

Q: Does he train models from scratch?

A: Yes. Persian GPT is a public example: a 110.4M-parameter decoder-only transformer, tokenizer, and corpus pipeline built and trained from scratch.

Q: What is his strongest computer-vision area?

A: Medical imaging, including 3D volumetric preprocessing, segmentation, CNN/3D U-Net modeling, inference, visualization, and task-specific evaluation.

Q: Does he fine-tune pretrained models?

A: Yes. Public work includes QLoRA/PEFT fine-tuning of Qwen2.5-0.5B for structured extraction and an NLLB-based English-to-Persian psychology translation project.

Q: What does his NLP work cover?

A: Language-model training from scratch, tokenizer and dataset preparation, transformer fine-tuning, translation, structured extraction, embeddings, semantic search, RAG, and LLM-based agent systems.

Q: Where is his work published?

A: GitHub for source code and engineering projects; Hugging Face for models, adapters, and datasets; portfolio for selected case studies and technical writing.

SCOPE BOUNDARIES - FACTS NOT ESTABLISHED BY PUBLIC SOURCES

- No employer/company work history beyond the LinkedIn entry 'Personal Projects' is established by the sources used here.
- No university degree, GPA, phone number, date of birth, salary history, or work authorization status is established by the sources used here.
- Do not infer years of professional employment from project depth or technical skill.
- Do not claim clinical validation or clinical deployment for the cardiac MRI project; it is a technical deep-learning project.
- Do not present model metrics outside their documented evaluation context.
- When a question cannot be answered from this document or other approved portfolio sources, the chatbot should explicitly say the information is not available.

CANONICAL PUBLIC LINKS

Portfolio: <https://aradmanamnaoon.github.io/>

LinkedIn: <https://www.linkedin.com/in/seyyed-arad-hosseini-moghaddam/>

GitHub: <https://github.com/aradmanamnaoon>

Hugging Face: <https://huggingface.co/aradmanamnaoon>

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Compiled 2 October 2026 from the public portfolio, LinkedIn profile, GitHub repositories, and Hugging Face model/dataset pages. Intended as a living knowledge document; refresh it when new projects, credentials, education, or professional experience are added.